

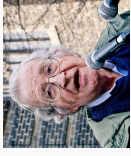
Overview of AI and NLP

Natural Language Processing, Week 1, Summer 2026

Dr. Paul Nulty

Lecturer
School of Computing and Mathematical Sciences
Birkbeck, University of London

How it started



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- (Noam Chomsky)

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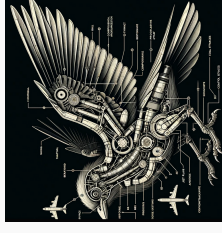
- 'The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin.'
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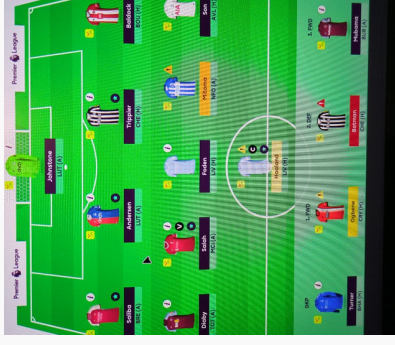


How its going — language

- General purpose intelligent language agents

How its going — language

- General purpose intelligent language agents
- Incredibly effective at answering creative linguistic and general intelligence tasks



How its going — language

On the bench, you have:

Goalkeeper: Turner

Defenders: Botman

Midfielders/Forwards: Ogbenne, Mubama

With this information, I can provide more targeted advice on potential transfers or strategy changes. Keep in mind that the effectiveness of these changes can be

How its going — language

PA

You

Here is the price list: A.Becker Liverpool 56 £5.6

Ederson M. Man City 37 £5.5

Pope Newcastle 47 £5.5

Vicario Spurs 47 £5.2

Arrizabalaga Chelsea 0 £5.0

José Sá Wolves 36 £5.0

Martinez Aston Villa 30 £4.9

Raya Arsenal 35 £4.9

Lloris Spurs 0 £4.8

Onana Man Utd 43 £4.8

Ramsdale Arsenal 11 £4.7

How its going — language



ChatGPT

Sure, let's adjust the team to maximize the budget while staying just under £101 million:

1. **Goalkeepers:** Keep Vicario (£5.2m) and Johnstone (£4.6m). Total: £9.8m.
2. **Defenders:** Retain Trippier (£7.0m), Andersen (£5.1m), and Saliba (£5.4m). Upgrade one defender to a mid-range option like Cash (Aston Villa, £5.2m), and include Tomiyasu (Arsenal, £4.5m) as your budget option. Total for defenders: approx. £27.2m.
3. **Midfielders:** Include Salah (Liverpool, £13.0m), Saka (Arsenal, £8.6m), and Bowen (West Ham, £7.7m). Keep Doucouré (Everton, £5.5m) and include a budget midfielder (approx. £5.0m). Total for midfielders: approx. £39.8m.
4. **Forwards:** Haaland (Man City, £14.0m) remains your premium forward. Keep Watkins (Aston Villa, £8.5m) and choose a forward like Wilson (Newcastle, £7.9m). Total for forwards: approx. £30.4m.

How its going — language

- Conversations with books

How its going — language

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- <https://goatgreatesteconomistofalltime.ai/en/chat/gpt>

GOAT: Who is the Greatest Economist of all Time and Why Does it Matter?

By Tyler Cowen



How did Keynes' views on inflation differ from Hayek's?



EconGOAT GPT-4:

Keynes and Hayek's views on inflation reflect their broader economic philosophies which were, in many ways, diametrically opposed.

Keynes viewed inflation as a possible consequence of an economy running near or at full capacity. He believed that in the short run, economic output is greatly influenced by aggregate demand. To Keynes, managing aggregate demand through fiscal and monetary policy could help mitigate economic downturns and stabilize inflation. Simply put, Keynes was not inherently adverse to inflation, especially if it was a byproduct of achieving lower unemployment. Keynes would argue that a moderate inflation rate can be a sign of a healthy, growing economy, and he was much more concerned about the dangers of deflation.

Hayek, on the other hand, approached inflation from a different angle. Firmly rooted in the Austrian School

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- Animal communication (natural, but not always considered to be language by linguists)

Related fields of study

- Computational Linguistics

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Modes of language

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- In linguistics, speech, writing and sign language are different realisations of human language
- No form is more perfect or more pure than another: they are all realisations of our underlying *faculty of language*

Design features of Language (Charles Hockett) - All communication systems:

- A mode of communication (e.g. vocal/auditory, gestural)

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- Discreteness: language is made up of discrete units (phonemes, morphemes, words, etc.)
- (discreteness is an important property when considering language as a computational object)



The subject matter of linguistics comprises all manifestations of human speech, whether that of savages or civilized nations, or of archaic, classical or decadent periods. In each period the linguist must consider not only correct speech and flowery language, but all other forms of expression as well.
Ferdinand de Saussure, Course in General Linguistics, 1916

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Subfields of linguistics

Table 1.1: A non-exhaustive sample of structural subfields of linguistics

Subfield	Description
Phonetics	The study of the sounds of human language
Phonology	The study of sound systems in human languages
Morphology	The study of the formation and internal structure of words
Syntax	The study of the formation and internal structure of sentences
Semantics	The study of the meaning of sentences
Pragmatics	The study of the way sentences with their semantic meanings are used for particular communicative goals

Phonetics

- Phonetics is a branch of linguistics that focuses on the production and classification of the world's speech sounds.¹

¹definitions in this section are from: <https://www.sheffield.ac.uk/linguistics/home/all-about-linguistics/about-website/branches-linguistics/phonetics>

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- Phonetics is a branch of linguistics that focuses on the production and classification of the world's speech sounds.¹
- Conventional orthography does not map well onto sounds

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Phonetics

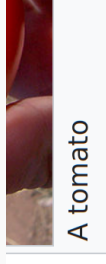
- Phonetics is a branch of linguistics that focuses on the production and classification of the world's speech sounds.¹
- Conventional orthography does not map well onto sounds
- **ghoti** is a creative respelling of the word fish, used to illustrate irregularities in English spelling and pronunciation.

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- *gh*, pronounced /f/ as in *enough* /ɪˈnʌf/ or *tough* /tʌf/;
- *o*, pronounced /ɪ/ as in *women* /ˈwɪmɪn/;
- *ti*, pronounced /ʃ/ as in *nation* /ˈneɪʃən/ or *motion* /ˈmoʊʃən/.

Pronunciation [\[edit \]](#)

- (*UK, General Australian*) **IPA**^(key): /
tə'mɑ:təʊ/



A tomato

- Audio (UK) [tʰə'mɑ:tʰəʊ]



- Audio (AU) [tʰə'mæ:tʰəʊ]



- (*US*) **IPA**^(key): /tə'meɪ.təʊ/

- Audio (US) [tʰə'meɪroʊ]



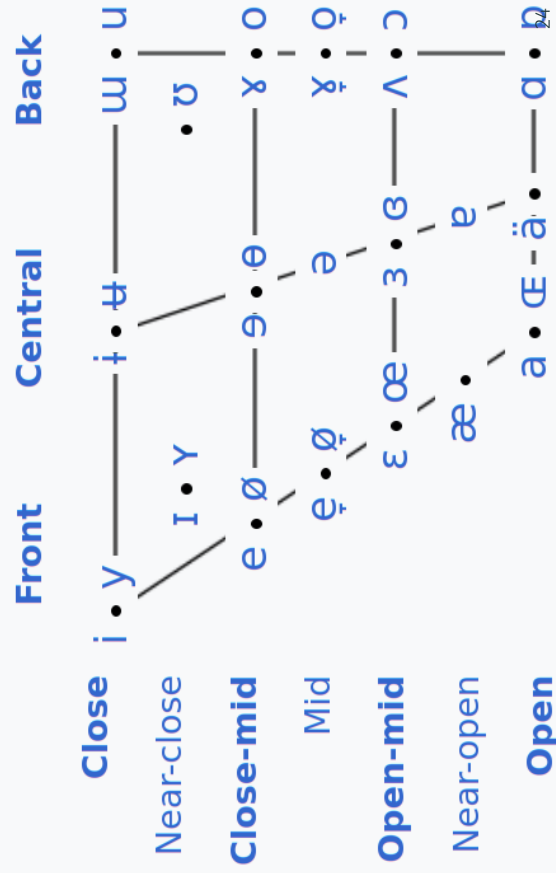
- (*Canada*) **IPA**^(key): [tʰə'meɪro], [tʰə'me(:)to(:)], [tʰə-], [-ma-]

the International Phonetic Alphabet

Consonants (pulmonic)

	Bilabial	Labiodental	Linguolabial	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular
Plosive	p b	p̥ b̥	t̪ d̪	t̪ d̪	t d		ʈ ɖ	c ɟ	k g	q
Nasal	m	m̥	ɱ	ɳ	n		ɳ	ɲ	ŋ	ɴ
Trill	ʙ		ɽ	ɽ	r					ʀ
Tap or Flap	ɾ	ɾ̥	ɽ	ɽ	ɾ		ɽ			ɽ̥
Lateral Tap or Flap					ɭ		ɭ	ʎ	ʎ	
Fricative	ɸ β	f v	θ ð	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ
Approximant	β̞	ʋ		ʋ	ɹ		ɻ	j	ɰ	
Lateral Fricative				ɬ ɮ	ɬ ɮ		ɬ ɮ	ɬ ɮ	ɬ ɮ	
Lateral Approximant			ɭ	ɭ	ɭ		ɭ	ɭ	ɭ	

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- allophones: different realisations of the same phoneme
- e.g. pin, bin, spin

Phonological Rules: phonemes and allophones

- I can see

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- I can see
- I can bake

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- inflectional bound morphemes modify the tense, aspect, gender, case

Affixation

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- There are rules!

Analytic and synthetic languages

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- Synthetic languages use bound morphemes: word order can be more flexible
- Agglutinating languages are highly synthetic and many aspects of meaning are expressed with bound morphemes

Morphology in NLP

- The goal of both stemming and lemmatization is to reduce inflectional forms and sometimes derivationally related forms of a word to a common base form. For instance:
 - am, are, is $\Rightarrow$ be
 - car, cars, car's, cars' $\Rightarrow$ car
- The result of this mapping of text will be something like:
 - the boy's cars are different colors $\Rightarrow$
 - the boy car be differ color

Syntax

- is the study of how words are ordered in human language. What rules are necessary to describe a particular language, and how can these rules differ across languages?
- In linguistics, syntax (/ˈsɪntæks/ SIN-taks) is the study of how words and morphemes combine to form larger units such as phrases and sentences. Central concerns of syntax include word order, grammatical relations, hierarchical sentence structure (constituency), agreement, the nature of crosslinguistic variation, and the relationship between form and meaning (semantics). ²

²<https://en.wikipedia.org/wiki/Syntax>

Word order

- The dog saw the man

Word order

- The dog saw the man
- The man saw the dog

Word order

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Word order

- The dog saw the man
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- *The man saw dog the
- ?Saw the man the dog

Constituents

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- Stand alone (e.g. as a response to a question)
- Replace with a pronoun
- Move together

Constituents

- constituents in a sentence that can be substituted are syntactic categories

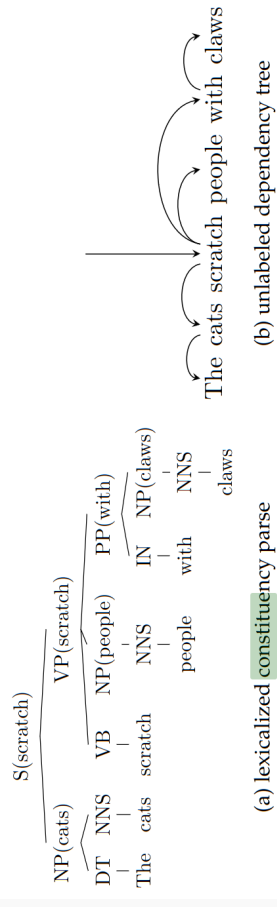
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- they are *parts of speech* (or are composed of them)

parsing



History of linguistics

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- 1600s-1800s focus on comparative and historical linguistics (philology)

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- Early 20th century: structuralism, the distributional hypothesis, Wittgenstein

1957: Syntactic Structures - Noam Chomsky



We can determine the adequacy of a linguistic theory by developing rigorously and precisely the form of grammar corresponding to the set of levels contained within this theory, and then investigating the possibility of constructing simple and revealing grammars of this form for natural languages.

1957: Syntactic Structures - Noam Chomsky



Despite the undeniable interest and importance of semantic and statistical studies of language, they appear to have no direct relevance to the problem of determining or characterizing the set of grammatical utterances. I think that we are forced to conclude that grammar is autonomous and independent of meaning, and that probabilistic models give no particular insight into some of the basic problems of syntactic structure.

Semantics

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- lexical semantics: study of word meaning.
- reference, truth conditions, structure of meaning

Pragmatics

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- pragmatics considers: the negotiation of meaning between speaker and listener.

and many more...

- psycholinguistics, sociolinguistics, neurolinguistics

Further Resources and discussion

- Dan Jurafsky and James H. Martin. Speech and Language Processing (3rd ed. draft). Chapters 1–9, 11, 20, 24, Appendix B.
- Pinker, S. (1995). The language instinct: The new science of language and mind. Penguin UK.
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Further Resources and discussion

- Schütze, H., Manning, C. D., & Raghavan, P. (2008). Introduction to information retrieval (Vol. 39, pp. 234-265). Cambridge: Cambridge University Press.
- lots of podcasts... [lingthusiasm](#), [Tomayto Tomahto](#) (including [Steven Pinker](#)), [Edward Gibson](#) ([Lex Fridman](#))
- Movies: [Arrival](#), [Project Nim](#), [Dreaming of Words](#)